

ANSWERS

1.

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1
5(b)(iv)	less diffraction or less bending	B1

2.

Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz)	A1
7(c)(iii)	1.3 (cm)	B1

3.

Question	Answer	Marks
8(a)(i)	3	B1
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	B1
	1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i)	B1
8(b)(i)	move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight	B1
8(b)(ii)	<i>speed</i> remains the same	B1
	<i>wavelength</i> decreases	B1
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary	B1
8(c)(ii)	diagram showing within the ripple tank:	B1
	crests parallel to bar hitting boundary with a boundary labelled	
	correct refraction of plane crests either towards or away from normal on the same side of the normal or no direction change if crests are initially parallel to boundary but with a wavelength change	B1
	different wavelength shown before and after refraction and constant before and after (by eye)	B1

4.

Question	Answer	Marks
10(a)(i)	3 (.0) cm	B1
10(a)(ii)	number of waves per second or $5 / 10$ or $f = 1 / T$	C1
	2(.0) Hz	A1
10(a)(iii)	$(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form	C1
	6(.0) cm / s	A1
10(a)(iv)	frequency stays the same and speed increases	B1
10(a)(v)	$\geq$ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1

5.

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	C1
	6.4–7.2 cm	A1
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	C1
	number of wavelengths/crests/troughs/ <u>wave</u> cycles/ <u>wave</u> oscillations generated/made/pass a point in one second/unit time	A1
	or number of oscillations of a <u>source/particle</u> in one second / unit time	
10(a)(iii)	2.5 Hz	B1
10(a)(iv)	moved up and down	B1
	regularly or at constant frequency / 2.5 times a second	B1
10(a)(v)	<i>speed</i> smaller / decreases and <i>wavelength</i> smaller / decreases	B1

6.

Question	Answer	Marks
9(a)	the <u>vibration</u> direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised	B1
9(b)(i)	$(f =) v / \lambda$ or $0.17 / 0.019$	C1
	8.9 Hz	A1
9(b)(ii)	(frequency) stays constant or does not change	B1
9(b)(iii)	(speed) decreases	B1
	(Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave	B1

7.

Question	Answer	Marks
11(a)(i)	tank with water	B1
	dipper (bar or point) touches or in surface of water	B1
	light source / stroboscope / video	B1
	dipper moves up and down (to create wave)	B1
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	B1
	moves up and down	B1
11(a)(iii)	line joining points	M1
	that are at the same point in the motion	A1
11(b)(i)	<i>difference</i> : Q has a greater / different amplitude or phase	B1
	<i>similarity</i> : same frequency; same time for one wave (period); both transverse; oscillation up and down	B1
11(b)(ii)	time 300 (ms) or 0.3 (s)	C1
	(f=) $1/t$ numerical or algebraic in any form	C1
	3.3 Hz	A1
11(b)(iii)	$2(\lambda =) v/f$ or $v = \lambda f$ – numerical or algebraic in any form	C1
	0.060 m	A1

8.

Question	Answer	Marks
6(a)	longitudinal – vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v/f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

9.

- (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
- water*: perpendicular **and** transverse B1
- (b) (i) 0.29 – 0.28 m B1
- (ii) time/period for one wave(length)/cycle constant B1
or each oscillation/cycle takes one second

10.

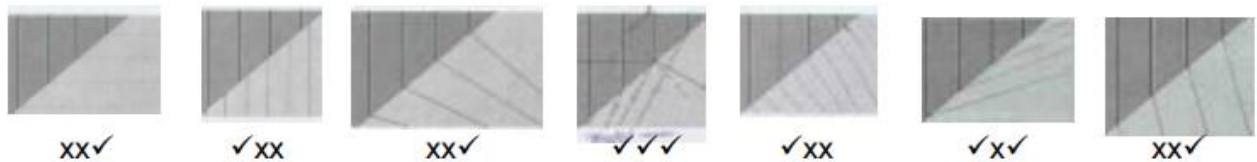
- (a) number of waves (that pass a point) M1
 or number of oscillations (passing a point)
 in unit time or per second or in 1 second A1

- (b) (i) 1.5 cm B1

- (ii) $(v =)f\lambda$ or 5×1.5 seen C1
 or $(s=)d/t$ and $f = 1/t$
 7.5 cm/s A1

- (c) (i) wavelength decreases B1
 travels a shorter distance in the same time B1
 or frequency stays the same (and $v = f\lambda$)

- (ii) wavefronts with smaller wavelength (by eye) B1
 smaller angle to surface (by eye) and slanted down B1
 wavefronts join those in shallow water B1



- (d) (i)

sound	water
particles / wave / source - vibrate / oscillate / move in direction of (travel of) wave / along wave - move backwards and forwards	particles / wave / source - vibrate / oscillate / move at 90° to direction of (travel of) wave - move up and down
(contains) compressions and rarefactions or particles come closer / further apart	(contains) crests and troughs
speed 300–330 m/s	wave slower (than sound)

B1
 B1
 one row only

- (ii) method of generating sound, e.g. (loud) speaker (and signal generator) B1
 apparatus that enables refraction clear, e.g. carbon dioxide in balloon or any shape where refraction is possible B1
 method of detecting refraction, e.g. microphone and how it is used to show refraction B1

11.		
(a)	oscillate / vibrate stated or described transverse movement described	B1 B1
(b)	0.40 m	B1
(c) (i)	$v = f \lambda$ or $(f =) v / \lambda$ or $2 / (b)$ 5.0 Hz	C1 A1
(ii)	clear attempt to draw wave moved along 0.20 m to right	B1
12.		
(a)	any two from: transmission of energy without net movement of medium through vibration of particles	B2
(b) (i)	number of (complete) waves / cycles / oscillations per unit time / second	B1
(ii)	distance between (neighbouring) waves distance between (neighbouring) wavefronts / points of same phase or crest to crest / trough to trough distance	C1 A1
(c)	three reflected wavefronts roughly correct direction wavelengths equal to each other and incident wavelength by eye reflected wavefronts joined to incident wavefronts	M1 A1 B1
13.		
(a) (i)	up and down clear, e.g. by double headed arrow or down	[B1]
(ii)	any correct distance between consecutive points in phase	[B1]
(iii)	correct distance	[B1]
(b)	measure number of oscillations / count waves (passing) in a stated time or time at least one oscillation show how to calculate number of oscillations per second	[B1] [B1]
(c)	moves (hand or rope) with slow(er) speed or rate / less frequency / less times per sec	[B1]

14.

- (a) (i) clear attempt at measuring more than one wavelength e.g. $6.85/5$
 or $1.30 - 1.45 \text{ cm}$ B1
 $10.7 - 11.3 \text{ cm}$ B1
- (ii) $(v =)f\lambda$ or $3.6 \times \text{(a)(i)}$ C1
 $40(39.6) \text{ cm/s}$ A1
- (b) (i) stays the same B1
- (ii) decreases B1

15.

- (a) Any 2 of
 • an oscillation/vibration/movement up and down
 • carries energy
 • no (net) movement of the medium/transfer of matter B2
- (b) arrow downwards or upwards or both B1
- (c) (i) $(v =)f\lambda$ in any form numerical or algebraic C1
 $5(.0) \text{ cm/s}$ or $0.05(0) \text{ m/s}$ A1
- (ii) line or indication **labelled D** of length 2 wavelengths B1

16.

- (a) $2(.0) \text{ mm}$ B1
- (b) same period (by eye), with at least one wave
 opposite phase to wave drawn B1
 B1
- (c) (i) $(f =)1/T$ numerical or algebraic seen (e.g. $1/0.5$) // 1 wave in 0.5 s // 2 waves
 in 1 s C1
 $2(.0) \text{ Hz}$ A1
- (ii) $v = f\lambda$ // 8×2 or $8 \times \text{(i)}$ // 16 (cm/s) // 5 (wavelengths from centre to edge) //
 $(t =)d/v$ C1
 2.5 s **ecf** from (i) – i.e. **accept 5/(c)(i)** A1

- 17.
- (a) at least 3 reflected wavefronts with same wavelength as before B1
all **and** ≥ 2 reflected wavefronts at correct angle B1
- (b) (i) no change in direction **and** clearly smaller, approximately constant wavelength B1
- (ii) 1 reduces B1
2 constant B1
- 18.
- (a) line joining points of same phase, e.g. line joining crests B1
- (b) (i) decreases B1
- (ii) decreases B1
- (iii) constant B1
- 19.
- (a) (i) up and down arrow shown
- (allow if one arrow and up/down stated in (ii)) B1
- (ii) 4 times in one second **or** once in 0.25 sec B1
- (b) $v = f\lambda$ in any algebraic form B1
- 0.8/4 C1
- 0.2 m A1
- (c) halve the frequency **or** move hand once every 0.5 springs B1
- (**ignore** move hand slower or at half speed or speed of wave double unless spring stretched more)